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And finally, I want to acknowledge the power of wireless capability and the Internet. While writing this book, I traveled north to Helsinki, Finland, and camped in Grande Prairie, Canada. I was south to Johannesburg in South Africa and camped in Botswana and Zimbabwe, writing between watching lions and elephants. As well I was in Australia, although I did not test wireless in the outback there. I even was as high as 3,000 meters (~10,000 feet) in Peru. There were only a few places where I could not connect at all. This writing was truly a distributed team effort.

Lisa's personal thanks:

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If readers learn a fraction of what I've learned while writing this book, I'll consider it a success!



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Parimala Hariprasad spent her youth studying people and philosophy. By the time she got to work, she was able to use those learnings to create awesome testers. She has worked as a tester for over ten years for domains such as customer relationship management, security, e-commerce, and health care. Her specialty is coaching and creating great teams—teams that ultimately fired her because she wasn't needed anymore. She has experienced the transition from web to mobile and emphasizes the need for design thinking in testing. She frequently rants on her blog, Curious Tester (<http://curioustester.blogspot.com>). She tweets at @CuriousTester and can be found on LinkedIn at <http://in.linkedin.com/in/parimalahariprasad>.

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Mike Heinrich has been working as a tester for over a decade, working in logistics, banking, telecommunications, travel, and utilities. Throughout his career, Mike has focused on data and integration testing. His passion for data and delivering customer value has afforded him the opportunity to present to a number of North American organizations on agile data warehousing and data testing. In his free time, Mike enjoys traveling the world, playing volleyball, and coaching basketball.

Sherry Heinze is a test strategist, tester, QA analyst, and trainer with a broad background in analysis, design, testing, training, implementation, documentation, and user support. For the last 17 years, Sherry has focused on testing from analysis and design forward, sometimes on cross-functional teams, sometimes with teams of testers, sometimes alone. Sherry has extensive experience working in various methodologies with both users and technical staff to identify and test requirements, design, create, test, implement, and support systems.

Matthew Heusser has spent his adult life developing, testing, and managing software projects. Along the way Matt served as a contributing editor for *Software Test & Quality Assurance* magazine, organized the Agile Alliance Sponsored Workshop on Technical Debt, and served on the board of directors for the Association for Software Testing. Perhaps best known for his writing, Matt was the lead editor for *How to Reduce the Cost of Software Testing* (Heusser, 2011) and is currently serving as managing editor for Stickyminds.com. As the managing consultant at Excelon Development, Matt manages key accounts for the company while also doing consulting and writing. You can read more about Matt at the Excelon website, www.xndev.com, or follow him on Twitter: @heusser.

Michael Hüttermann, a Java champion, is a freelance delivery engineer and expert for DevOps, continuous delivery, and source control management/application life cycle management. He is the author of *Agile ALM* (Hüttermann, 2011a) and *DevOps for Developers* (Hüttermann, 2012). For more information see <http://huettermann.net>.

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Stephan Kämper studied physics, wrote his diploma thesis about holography, and then joined the oceanography group at the University of Bremen. In 2001 he started in software development by joining the test team for an object-oriented database system. He never left software testing and specialized in automated software tests and agile methods. He worked on topics as diverse as precision navigation systems, payment platforms, health care systems, telecommunication, and social networks. Working in these different fields helped him recognize common patterns, which he found useful in software testing. His languages are (in alphabetical order) English, German, and Ruby. Follow him on Twitter at @S_2K, and see his website: www.seasidetesting.com.

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Claire Moss became the first discrete mathematics business graduate from the Georgia Institute of Technology in 2003 and immediately jumped into software testing. She has been following this calling ever since, working with agile product teams as a testing teacher, unit and integration test adviser, exploratory tester, and test automator. Although she'll always go back to scrapbooking, her dominant hobby in recent years has been writing, speaking, and nerding about testing. Claire has always had a passion for writing, and she continues to use her evil powers for good on the job and on her blog at <http://aclairefication.com>.

Aldo Rall started off in testing as a junior programmer at the start of the Y2K bubble. Since then, working in South Africa and the UK, he gained practical experience in testing across a plethora of titles, assignments,

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Sharon Robson is the software testing practice lead for Software Education. A passionate tester and a natural-born trainer, Sharon delivers and develops courses at all levels of software testing from introductory to advanced. Sharon also focuses on agile and spends a significant amount of her time working with teams (training, coaching, and mentoring) to assist them in their transitions. Sharon is currently researching and writing about agile test approaches in various business domains. She presents at both local and international conferences and contributes to the testing and agile community via blogs, tweets, conference involvement, and mentoring.

Steve Rogalsky, recognizing that software development culture, management, and process can be frustrating and inhibiting, has invested significantly in finding ways to overcome and counteract those effects. He’s found that valuing simplicity, respect for people, continuous improvement, and short feedback loops are powerful tools for addressing these shortcomings. Since software development doesn’t own those frustrations, he’s also been translating what he’s learned into other areas of the organization, family life, community groups, and coaching. He speaks regularly at conferences in Canada and the United States, has been featured on InfoQ, cofounded the Winnipeg Agile User Group, and works at Protegra. You can read more about what he’s learned at <http://WinnipegAgilist.Blogspot.com>.

Bernice Niel Ruhland, with over twenty years of professional experience encompassing a variety of technical disciplines, currently serves as the director of quality management programs for ValueCentric LLC. Applying her proficiencies in software programming, testing, assessment, and implementation, Bernice leads the company’s software testing department. As the driving force behind ValueCentric’s company-wide quality programs, she draws upon practices in the context-driven and agile theories and methodologies to guide foundational efforts. When not working, she maintains a successful blog, www.TheTestersEdge.com, a collection of her observations related to a variety of technical topics including software testing, leadership, and career development.

Huib Schoots is a tester, consultant, and people lover. He shares his passion for testing through coaching, training, and giving presentations on a variety of test subjects. Curious and passionate, he is an agile and context-driven tester who attempts to read everything ever published on software testing. He's also a member of TestNet, AST, and ISST; a black belt in the Miagi-Do School of Software Testing; and coauthor of a book about the future of software testing. Huib maintains a blog on www.magnificant.com and tweets as @huibschoots.

Paul Shannon and **Chris O'Dell** joined the 7digital team in 2010 and 2011 respectively, both starting in the team responsible for the 7digital API. They worked on improving the quality of the testing in the API, and Chris now leads that team, concentrating on improving the platform for continuous delivery, resilience, and scaling. Paul works across all teams in the 7digital technology team that are geared toward continuous improvement and quality-driven software development practices. The team follows a test-first approach with a highly collaborative and visible workflow, and all absolutely love technology and testing.

Jennifer Sinclair has been an artist, art instructor, and educator since 1995. During that time, she has lived in Canada, Japan, and the United States and has worked to improve art exploration for children and adults of all ages and abilities. She has designed and illustrated images for the Alberta Teachers' Association Early Childhood Education Council and the Alberta Education Council in Canada. She is currently working on developing art lessons that integrate easily into the core subjects of elementary education. As a homemaker, freelance artist, and volunteer art instructor, she is passionate about continuing to develop her skills and knowledge and share them with as many people as possible. You can reach her at jvaagesinclair@live.com.

Toby Sinclair joined the software testing business as a university graduate in 2007 and hasn't looked back. He has worked for various software testing consultancies in the UK and is currently working with J. P. Morgan to advance its testing capabilities to support the transition to agile. Toby is an active member of the testing community and can be found on Twitter: @TobyTheTester.

Tony Sweets is a 20-year veteran of the software industry, currently working as an information technology architect. For the past 13 years

he has been working on Java enterprise web applications in the financial sector. Tony possesses a wide range of skills but likes to work mostly on Java applications and the tools that make the development process better. Tony holds a bachelor's degree in computer science from the University of Wyoming.

Mike Talks was 26 when he first gave IT “a go” as a career. Before that, he'd been a teacher, research scientist, and data analyst, and his parents worried he'd never “get a proper job.” Although originally a programmer, it was in testing where he flourished. He originally worked on long, requirements-rich military waterfall projects in the UK, but since moving to New Zealand he's found himself increasingly working on projects with companies such as Assurity, Kiwibank, and Datacom, where timely delivery is a key factor.

Eveliina Vuolli acts currently as operational development manager in Nokia Solutions and Networks. She has been working with the network management system R&D development team for 15 years, acting in different kinds of roles in the global, multinational organization: integration and verification process owner, project manager and trainer in various areas, and also coach. In addition, she has been involved in the agile transformation in her own product area.

Pete Walen has been in software development for over twenty-five years. He has worked in a variety of roles including developer, business analyst, and project manager. He is an independent consulting contractor who works with test teams over extended periods, coaching them and working to improve their testing techniques and practices. Pete describes himself as a “Software Anthropologist and Tester,” which encompasses the examination of how software and people relate and interact. He has worked in a variety of shops using a variety of development methodologies and has adopted an attitude of “do what makes sense” for the organization and the project.

Mary Walshe helps teams deliver successful solutions to business problems and plays a major role in striving for a kaizen culture in these teams. Mary was the driving force behind the introduction of acceptance-test-driven development in her department. She has been

working in the industry for four years, and currently she works on a team in Paddy Power as an agile tester. Her team is using kanban to help them measure their experiments and in order to continually improve. In her spare time Mary runs adventure races, mountain bikes, and just recently found a new love for skiing.

Christin Wiedemann, after finishing her Ph.D. in physics at Stockholm University in 2007, began working as a software developer. Christin soon discovered that she found testing to be more challenging and creative, and she joined the testing company AddQ Consulting. There, she worked as a tester, test lead, and trainer, giving courses on agile testing, test design, and exploratory testing. In late 2011, Christin moved to Vancouver, Canada, joining Professional Quality Assurance. In her roles as tester, test lead, trainer, and speaker, Christin uses her scientific background and pedagogic abilities to continually develop her own skills and those of others.

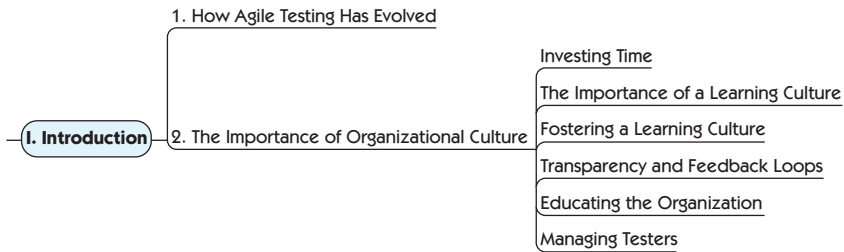
Lynn Winterboer, with a proven background in a variety of data projects and agile practices, teaches and coaches data warehouse/business intelligence teams on how to effectively apply agile principles and practices to their work. For more than fifteen years, Lynn has served in numerous roles within the analytics, business intelligence, and data warehousing space. She very well understands the unique set of challenges faced by teams in this area that want to benefit from the incremental style of agile development; Lynn leverages her experience and training to help deliver practical solutions for her clients and students. Lynn can be reached at www.LynnWinterboer.com.

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INTRODUCTION

In this first part, we go through a brief history of how agile testing has changed since we first worked on agile teams. We introduce some of the new concepts and go into more detail about the importance of a learning organizational culture.



- **Chapter 1**, “How Agile Testing Has Evolved”
- **Chapter 2**, “The Importance of Organizational Culture”

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Chapter 1

HOW AGILE TESTING HAS EVOLVED

Each of us started our “agile” career as the lone tester on an Extreme Programming (XP) team. At the time, none of the publications about XP even mentioned having testers on the team. There were two roles: programmer and customer. Customers would specify all the acceptance tests, programmers would automate them, and presto, you’re done. When we went to XP conferences, we were usually the only people attending who self-identified as testers, although we knew that testers had a lot of value to add. We experimented, we discussed testing with the XP pioneers, and we traded ideas with each other and with other members of our respective teams.



Driving
development
with examples

Agile development evolved, and agile testing evolved along with it. Teams started developing test libraries and frameworks for test automation above the unit level. By the time we wrote *Agile Testing*, many agile practitioners acknowledged the contributions of experienced testers such as Elisabeth Hendrickson and Michael Bolton. Practitioners such as Brian Marick and Joshua Kerievsky had pioneered the idea of using examples and story tests to guide development.

Ward Cunningham created Fit (Framework for Integrated Test), a tool to help define those examples, and Dan North introduced behavior-driven development (BDD) (North, 2006), which paved the way for popular new tools. Agile teams had begun to understand the value of exploratory testing. Testing on agile teams had gone beyond the functional to embrace many types of testing, as illustrated by Brian Marick’s agile testing matrix (Marick, 2003), which we adapted into the agile testing quadrants (we also refer to them as the Quadrants).

Of course, there were still challenges that hindered agile testing success. We testers envied all the aids for unit testing that were built into the programmers' integrated development environments (IDEs). We wanted that same ease for testers specifying tests. We found huge benefits in applying the "Power of Three," or "Three Amigos," as George Dinwiddie calls it (Dinwiddie, 2010), where a customer, programmer, and tester collaborate anytime there are questions about how a feature should behave. However, we still found it difficult to capture many dimensions of customer requirements such as design, usability, data, and other quality attributes. These are some of the challenges we address in this book.

Practitioners in several disciplines have been filling in gaps in the practice of agile testing. We're fortunate to be able to share other practitioners' stories about how they have succeeded with agile testing in different contexts.

One key idea has become part of our everyday thinking: testing in agile teams is an activity, not a phase. We learned this concept from Elisabeth Hendrickson (Hendrickson, 2006) and apply it throughout the book to emphasize that testing is something that all disciplines need to consider while developing software.



Continual
learning

As we've continually learned more and better techniques for testing in agile, we've discovered how generalizing specialists with both depth and breadth in their skill sets help teams tackle difficult testing challenges. Practitioners have created practices and patterns that help team members in different disciplines collaborate and learn "just enough" of the specialties from each other.

Practitioners such as the members of the Agile Alliance Functional Test Tools (AA-FTT) group have led the way to better tooling for testing. Today, writing test code is on a par with writing production code. We've learned better ways to identify which frameworks, test libraries, and drivers are a good fit for a specific team's needs.

Business analysts have brought their specialized skills at discovering customer requirements into agile development. Testing and business analysis share complementary skills that help teams deliver the right business value. Similarly, user experience (UX) experts have shown us

good, simple ways to get customer input as new features are designed. DevOps practitioners have blended development, operations, and testing skills to advance quality in new dimensions such as delivery and deployment and to help shorten release cycles to reduce risk and the impact to customers.

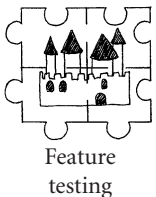


We continue to meet many more agile teams that understand the value of exploratory testing than we saw a few years ago. Although our first book covers exploratory testing, it's clear that teams can take advantage of its power in many more ways. We're fortunate that people who practice exploratory testing in an agile context are sharing their expertise.

We've seen new, creative approaches to test planning and collaboration. Teams today find more ways to visualize quality. The Quadrants and the test automation pyramid have been adapted and extended to represent more dimensions.



More teams today are faced with testing mobile apps and embedded software, on an ever-expanding range of devices and platforms. Large and complex data sets with new technology to store, manage, analyze, search, and visualize the data define a new category: Big Data. Testing has to keep up.



Agile development arose for use in small, colocated teams. Now we see that large organizations, some of which started as small agile companies, as well as distributed teams, are using an agile approach to development. In organizations with enterprise-wide software solutions, testing runs into different obstacles, such as heavyweight organizational constraints. At the same time, organizations are finding new ways to develop minimum viable products (MVPs), using iterative releases with fast feedback loops and validated learning.

While some companies may have been testing in production back in 2008 when we wrote our first book, we didn't start hearing about that technique until later—unless you count releasing without testing and hoping for the best! Today, releasing updates to a small percentage of production users while monitoring log files for errors, and other techniques of getting fast feedback from production users, can be a valid and necessary strategy in the appropriate context.



Keeping
it real

All these changes and innovations prompted us to share what we and other practitioners have learned. Agile is about continually improving, and many of you may now be several years into an agile adoption, changing your process as you learn. The testing-related problems you face now are likely much different from what they were during your initial transition to agile. We hope the experiences shared in this book will give you some ideas for experiments to try as you visualize quality for your product and then inspect and adapt your process.

SUMMARY

As agile software development evolves, we continually inspect and adapt agile testing to keep pace. In this chapter, we looked at some ways agile testing has been changing.

- We've recognized that testing is an activity that is crucial to product success and is of interest to every member of the product team from stakeholders right through to operations and support.
- Agile teams increasingly recognize the ways practitioners from other disciplines such as business analysis, user experience design, and DevOps have expanded our ability to build in the quality our customers desire.
- Tooling for agile testing continues to make great strides forward, enabling agile teams to implement the infrastructure needed to support learning and quick feedback.
- Agile teams are learning the value of exploratory testing and other practices that help us provide essential information to customers and developer teams.
- Agile testing has to keep up with fast-changing technology and new contexts for agile development. We'll explore these ideas throughout this book.

THE IMPORTANCE OF ORGANIZATIONAL CULTURE

2. The Importance of Organizational Culture

Investing Time

The Importance of a Learning Culture

Fostering a Learning Culture

Transparency and Feedback Loops

Educating the Organization

Managing Testers

We've noticed a common theme in organizations that are transitioning to agile methods. Development teams are able to adopt agile values, principles, and practices. However, the business side of the company may be slow to understand the benefits of agile and may often take longer to adjust to the change. For a successful transition to agile, it is important that both development and business be in sync and understand each other's reality.

When you work in a company whose leaders understand that focusing on quality is one of the best ways to deliver business value to customers frequently, your team—and company—are likely to succeed. Frequency and consistency of delivery improve over time. Conversely, if a company's focus is on speed first, often quality is sacrificed. Technical debt in the form of fragile, hard-to-change code and slower feedback cycles decreases the team's ability to deliver consistently.

If teams feel pressured to meet unrealistic deadlines with a defined scope, they're no longer in control of their own workload. As the deadline looms close, developers feel they have fewer options and make choices to fit the deadline rather than satisfy the customer (Rogalsky, 2012). They're likely to start cutting corners and make decisions



Figure 2-1 Too many features for the infrastructure

without thinking of potential ripple effects, and they may fall into a death spiral of ever-growing technical debt.



Teams need to be able to articulate the dangers of an unsustainable pace so that company leaders are aware of the risks and the consequences. Sometimes a team needs to slow down to address technical debt, or maybe to learn new concepts or apply new ideas. Janet uses a picture (Figure 2-1) in her presentations to show how adding more and more features without considering the infrastructure can actually be detrimental. The picture shows four superpowered cars sitting in a lineup on a road—going nowhere because the traffic is so heavy. Why add more features when the ones you have don't work as intended? When you build up too much technical debt, your team suffers a similar state of gridlock.

INVESTING TIME

Creativity is required to give customers useful software solutions, and people need time to think and use their ingenuity. We also must keep learning new technical skills and tools in order to conduct small experiments that might help address their problems.

In an article titled “Slow Ideas” (Gawande, 2013) Atul Gawande explored why some innovations, such as surgical anesthesia, catch on quickly, while others, such as antiseptics, take many years or may never get traction. His observation was that change can occur only with learning, practice, and a personal touch. In the programs he studied, people learned best when they performed the new activity themselves and described it in their own words, with guidance from a trainer. It may seem cheaper or faster to have a trainer demonstrate the activity or have people watch a video, but the hands-on approach produces real, sustainable change.



In the software profession (as in many others), people seem to forget that it takes time to learn and practice a new skill. They also often neglect to have that trainer alongside to guide the learners.

Spend Time on the Right Thing

David Evans, *an agile quality coach, shares a metaphor he uses to explain the idea of spending time on the right thing.*

I sometimes get management resistance to my suggestions to invest time in collaboratively writing acceptance tests before implementing each story. The usual argument is that if the team spends more time on acceptance tests, it will slow down the rate of development. I concede that yes, in a very limited sense, it will slow down, but only in the same way that stopping for passengers slows down the public bus. Trying to optimize the speed of the bus is missing the point, since it has nothing to do with the purpose or success of public transport. Imagine a bus driver suggesting that he could improve his performance metrics if he didn't stop for elderly passengers, as they seem to be the slowest to board. In fact, if he didn't pick up any passengers at all, he'd be much faster and wouldn't have to stop at all those bus stops on his winding route. Instead, he could get on the expressway and be back in the depot in no time at all!

This is the same mistaken logic that says that working on tests will slow down development. In *Test-Driven Development: By Example*, Kent Beck has a nice statement (Beck, 2002): “Code that isn't tested doesn't work—this seems to be the safe assumption.” If spending time creating acceptance tests before coding slows anything down, it is only slowing the rate at which we are creating code that doesn't work.

Manufacturing companies that embrace lean values have taught us that respecting individuals' ability to contribute ideas as well as working continually to improve translate into innovative products that customers love. When organizations give teams time and support to learn and experiment, those teams can do their best work. In Part II, "Learning for Better Testing," we will explore some of the practices we think merit an investment in learning.

Gojko Adzic (Adzic, 2013) has noted that great results happen when

- People know **why** they are doing their work
- Organizations focus on delivering **outcomes and impacts** rather than features
- Teams decide what to do next based on **immediate and direct feedback** from the use of their work
- Everyone **cares**

Some companies have tried to create slack and give employees time to learn by building in slack time to work on personal projects, either as a percentage of work time or dedicated "hack" weeks or days. Lisa has worked on teams that use this practice, and they experienced good results in keeping technical debt low and throughput consistent. However, it's important to understand the principles behind utilization and capacity. Otherwise, teams might face trying to squeeze even more work in between hard deadlines.

Capacity Utilization

Alexei Zhiglov, a consultant in lean and kanban for knowledge-work organizations, explained the theory behind capacity utilization and how to escape the pitfalls of 20% time. We've included a portion of his sidebar here; the complete one can be found on the book's website.

The main reason for 20% time is to keep the average capacity utilization at 80% rather than at 100% to build in slack time. We can think of a software development organization as a system that turns feature requests into developed features. We can then model its behavior using queuing theory.

Theory

If requests arrive faster than the system can service them, they queue up. When arrivals are slower, the queue size decreases. Because the arrival and service processes are random, the queue size changes randomly with time (Figure 2-2).

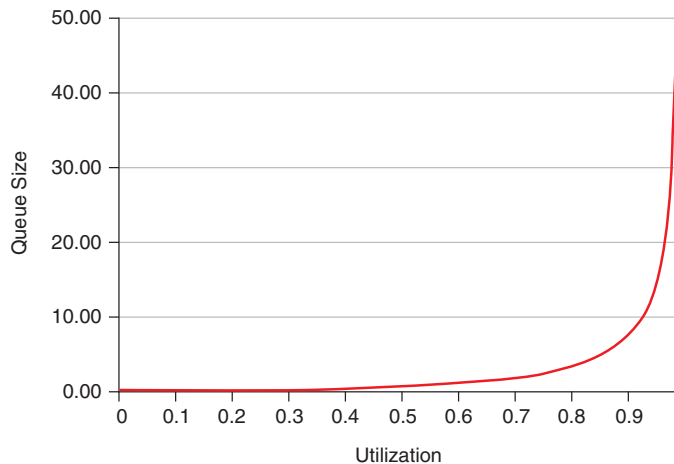


Figure 2-2 J-curve on utilization

The average queue size remains low while utilization is up to 0.8, then rises sharply and goes to infinity. We can understand this intuitively by thinking about our computer's CPU: when its utilization is low, it responds instantly to our inputs, but when a background task pushes its utilization close to 100%, the computer becomes frustratingly slow to respond to every click.

Practice

It is essentially the slack that keeps the system responsive. Several practical conclusions about what not to do are

- Cost accounting (engineers' time costs X , but the company may not be able to afford it). The economic benefit comes from reducing the cost of delay.
- Setting up a proposal system for people to submit projects to do in their "20% time."
- Tracking the 20% time by filling out time sheets.

- Using innovation as a motivator for the 20% time. While new products have come out of 20% projects, they were not the point. If your company cannot innovate during its core hours, that's a problem!
- Relying on the 20% time to encourage creativity. Saying you'll unleash your creativity with 20% time begs the question of why you're not creative enough already during your core hours.
- Allocating the 20% time to a Friday every week.

Those Are All Don'ts; Where Are the Dos?

OK, what about doing it right? Let's answer with the best question we've heard while discussing this subject with practitioners: "If 20% of your capacity is mandated to be filled with non-queue items, then you've just shrunk your capacity to 80%, and 80% is your new 100%. Right?"

Yes, "80 is the new 100" highlights the main problem with the attempts to mandate the 20% time without understanding the theory. You want to escape the utilization trap, not to stay in the trap and allocate time differently.

Utilization depends on two processes: arrival process (demand) and service process (capability). You can't really choose your utilization. It is what it is because the processes are what they are. You can, however, work on the processes by improving your company's software delivery capability and shaping the demand. As you make progress, slack will emerge.

The bibliography for Part I, "Introduction," includes resources where you can learn more about managing capability and throughput. One of the most important things to remember is that overutilization impacts flow, and unsustainable pace actually reduces throughput.

THE IMPORTANCE OF A LEARNING CULTURE

If you can "fail fast" enough so that the failure isn't too costly, you can learn from those mistakes to improve. You can also use fast feedback loops to refine your trials to change something that worked "OK" into something that works great. If your organization's culture is such that mistakes aren't tolerated, innovation is unlikely. Doing the same process, even if it's not working, feels safe, so there is no reason to change.

Lisa's Story

I've been fortunate to spend most of my career on teams that worked at a sustainable pace and valued continual improvement. When I've encountered environments where questions and experiments are discouraged, it feels like I've slammed into a brick wall. On one team I worked with for a short time, a ScrumMaster told me that I was an "impediment to the team" because I kept raising issues at standup. He was under pressure from management to get the team to deliver. He didn't want to be slowed down by someone pointing out that there had been no test environment for three weeks. There were lots of great people on the team, but they weren't allowed to do their best work. My attempts to be a change agent in this organization failed, so, as they say, I changed my organization.

In her talk on the power of retrospectives (Rising, 2009), Linda Rising pointed out that often, when confronted with practices such as pair programming or retrospectives, some managers have this reaction: "We don't have time for thinking." If we believe we can improve, we will—perhaps not all at once, or overnight, but we can find better ways to work, one baby step at a time.

FOSTERING A LEARNING CULTURE



Get your team together to see what skills might be missing and what people are interested in learning. Let them manage their workload so they can build in time for experiments, reading, training courses, and practicing new skills. Team members should feel empowered to research a new technique or try out a new tool. For example, they could be allowed to devote some time daily or weekly to work on anything they like, whether it's learning facilitation skills or finding a new solution for automating regression tests. Hack weeks where team members work on experimental projects or learn a new technology are an increasingly popular approach. Team members share what they've learned, helping to foster a learning culture.

Gojko Adzic told us that he has had success working with management and asking them if they expect the teams to improve their process over time, or if they think the process is already the best it can be. Having established that the process could indeed be improved, he works with them to develop a budget, not necessarily in terms of money, but instead